Energy from Water

Exam code: 0625 & 0972

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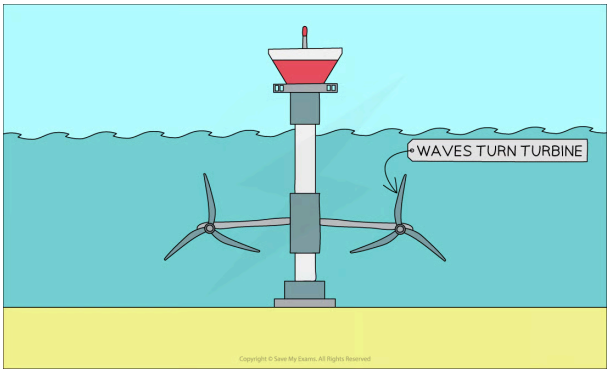
Wave & tide power

- Energy can be obtained from the kinetic store of the waves and tides
- The motion of the water is used to turn a turbine, which turns a generator, which generates electricity

Wave power

- Wave power uses underwater turbines
- The waves move the water which causes the turbines to turn

Turbines using wave power

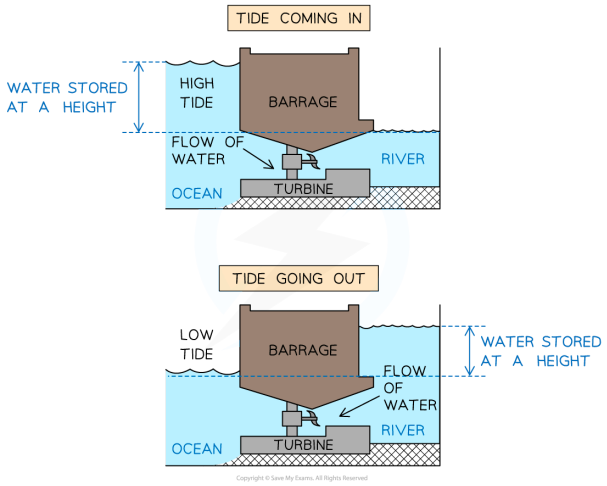


Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Tidal power uses tidal barrages to store water at a height as the tide comes in and goes out
- When the barrage is opened, water flows through a channel over a turbine

Tidal barrage



A tidal barrage stores water at a height and then opens a gate allowing water to flow over a turbine

Advantages of wave and tidal power

- Wave and tide power produce **no pollution** or greenhouse gases as no fuel is needed
- No fuel costs
- Tides are regular, so **tide power** is **reliable** and predictable
- **Large** amounts of electricity can be produced at **regular** intervals
- Wave and tidal power stations are **responsive**; they can be turned off and on quickly to meet demand
- Wave and tidal power are **renewable** energy resources
- Smaller power stations are are useful for **small islands**
- Wave and tidal power stations are **inexpensive** to build and cheap to run
- Wave and tidal power stations are **very efficient** (around 80%)

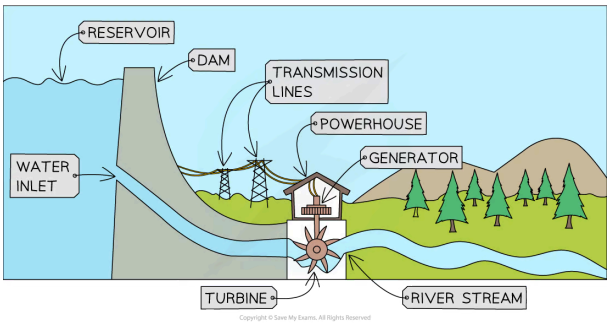
Disadvantages of wave and tidal power

- Tidal barrages are built in **estuaries** so there are not many suitable locations
- The underwater turbines and tidal barrages can **interfere** with shipping routes and cause **damage** to habitats
- The technology for wave and tidal power is not currently advanced enough for large-scale electricity generation
- Wave and tidal power stations have a **long construction time** compared to some other power station types

Hydroelectric dams

- When water is stored above ground level, it has energy in its **gravitational potential store**
- This energy can be transferred to **kinetic energy** if the water is allowed to flow down the slope
- Flowing water turns the turbine to **generate electricity**

A hydroelectric dam



A hydroelectric dam transfers energy mechanically from the gravitational potential energy store of the water to its kinetic energy store. The moving water turns the turbine

Advantages of hydroelectric power

- Hydroelectric dams are **reliable** and well established in most countries
- Hydroelectric power stations are **responsive** because they can be turned on and off easily to meet changing demand
- Hydroelectric power stations are cheap to run
- Hydroelectric power stations can produce **large** amounts of electricity
- Water is a **renewable** energy resource

Disadvantages of hydroelectric power

- Valleys are flooded to create the reservoir used to store the water, which **destroys habitats**, towns and villages
- **Pumping systems** are sometimes used to return the water to the reservoir
  - Pumping systems use **fossil fuels**, which release greenhouse gases and pollution into the atmosphere
- Hydroelectric power stations are **expensive to build** because the land and properties need to be acquired in order to build the reservoir
- High-voltage power lines are required to transmit electricity to the National Grid
- **Drought** can affect the supply of water

In your IGCSE exam, you may be required to discuss the advantages and disadvantages of various energy resources. You may also need to apply your understanding of energy resources to unfamiliar scenarios, for example, by suggesting which type of energy resource would be best for a particular community. Make sure you can justify your choices.

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